



DVM S 3 Phase Water Heat Pump
Heating Target Operation

Water Source Outdoor Unit				
Component	Abbreviation		Heating Mode	Comments
Compressor	Comp	Operation	Target High pressure control (default: 406 psig)	Primary function is to maintain head pressure (range 356-455psig)
		Range	20~160Hz	
Main EEV	E_M	Operation	HR - Target SH = Water temperature - 5.4°F	Valve modulates to maintain Evaporator Superheat
		Range	125 - 2000 Steps	Suction Superheat 1.8°F - 12.6°F
EVI EEV	E_EVI	Operation	On (Open)	By-Pass Mode 18°F Superheat Between pipe in and pipe out
		Range	0-480step	EVI Vapor Injection Mode 3.6°F Superheat Between pipe in and pipe out
4 Way Valve	V_4W	Operation	On (Energized)	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_W > 50.4 or 50Hz↓	By-pass mode, Discharge Temperature control
			Off (Close): T_W < 45°F ↓ and 65Hz↑	Low Ambient heating Operation Vapor Injection circuit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	
Liquid Line Solenoid Valve	V_L	Operation	On (Open)	
Main Cooling Solenoid	V_MC	Operation	Off (Closed)	
Hot Gas Charging Solenoid Valve	V_HC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 2 degrees lower than the water out temperature	Low Side saturation temperature is <i>"typically"</i> 2-10 degrees lower than water temperature
Indoor Unit IDU				
Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	61-86 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub_cooling Control	Used to calculate Evaporator Sub-Cooling (91°F - 122°F)
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil (113°F - 194°F)
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	